

FRONTEND ASSIGNMENT - 2

1. What is CSS Box Model?

The CSS Box Model is a foundation of layout design where every HTML element is treated as a rectangular box. It controls element dimensions and spacing through 4 layers:

- **Content:** The actual text, image, or media.
- **Padding:** Space inside the border around the content.
- **Border:** A line wrapping around the padding and content.
- **Margin:** External space outside the border that pushes other elements away.

2. What is media query? Explain with a sample code.

A media query (`@media`) is a CSS rule used in responsive design to apply custom styles based on device characteristics like screen width.

```
/* Normal view for desktop */
body {
    font-size: 16px;
    background: #fff;
}

/* Switches background & text size on screens 600px wide or smaller
*/
@media screen and (max-width: 600px) {
    body {
        font-size: 14px;
        background: #f5f5f5;
    }
}
```

3. What are the features of Bootstrap?

- **12-Column Grid:** Simple layout system that adjusts to screen sizes.
- **Pre-designed Components:** Ready-made components like navbars, buttons, modals, and cards.

- **Mobile-First Approach:** Default styles are built for smaller screens first and scaled up.
- **Utility Classes:** Fast helper classes for spacing, flexbox, alignment, and colors.
- **Cross-Browser Setup:** Handles browser inconsistencies out of the box.

4. Write an example of basic grid structure in Bootstrap.

Bootstrap uses a container-row-column structure:

```
<div class="container">
  <div class="row">
    <!-- Spans 6 columns on medium+ screens, 12 on mobile -->
    <div class="col-md-6 col-12">Box 1</div>
    <div class="col-md-6 col-12">Box 2</div>
  </div>
</div>
```

5. What are CSS Selectors?

CSS selectors target specific HTML elements so you can style them.

- **Element Selector:** Targets HTML tags directly (`p { color: black; }`).
- **Class Selector:** Targets elements with a class name (`.card { padding: 10px; }`).
- **ID Selector:** Targets a single unique ID (`#header { margin: 0; }`).
- **Universal Selector:** Targets every element on the page (`* { box-sizing: border-box; }`).

6. What are CSS backgrounds, list the properties?

CSS background properties style the surface area behind element contents:

- `background-color` : Sets a solid background color.
- `background-image` : Applies an image as the background.
- `background-repeat` : Prevents or controls image tiling (`no-repeat` , `repeat-x`).
- `background-position` : Sets where the background image starts (`center` , `top left`).
- `background-size` : Controls image scaling (`cover` , `contain`).
- `background-attachment` : Keeps the image fixed or lets it scroll with the page (`fixed` , `scroll`).

7. How can we overlap elements in CSS?

- **Positioning + z-index** : Set `position: absolute` or `relative` on elements to shift them, then use `z-index` to decide which element stays on top.
- **Negative Margins**: Use negative values like `margin-top: -20px` to pull an element over an adjacent one.
- **CSS Grid**: Place multiple items inside the exact same `grid-area` or row/column coordinates.

8. What are the different positioning properties in CSS?

- **static** : The natural page flow order (default).
- **relative** : Offset relative to where it normally sits without breaking page layout.
- **absolute** : Taken out of normal flow and positioned relative to its closest non-static parent.
- **fixed** : Locked to the screen viewport; stays in place during scrolling.
- **sticky** : Scrolls like `relative` until reaching a set scroll point, where it locks like `fixed`.

9. What is ECMAScript?

ECMAScript is the standard specification that defines the rules, syntax, and features for scripting languages. JavaScript is an actual implementation of ECMAScript.

Updates like ES6 introduced modern features like block scoping, arrow functions, and classes.

10. What is the difference between var, let and const in JavaScript. Explain Temporal Dead Zone?

- **var** : Function-scoped, can be redeclared and reassigned, hoisted as `undefined`.
- **let** : Block-scoped, cannot be redeclared in the same scope, can be reassigned.
- **const** : Block-scoped, cannot be redeclared or reassigned.

Temporal Dead Zone (TDZ): The TDZ is the period between the start of a block scope and the line where a `let` or `const` variable is declared and initialized. The variable is hoisted, but trying to read or modify it inside this zone throws a `ReferenceError`.